

Gaussian Process Multi-fidelity Modeling of Near-Earth Galactic Cosmic Ray Spectra

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1 **2026-06-25** *Radial Basis Functions: fixed and parametric*

1.1 The general idea

2 2026-06-25 *Kriging: A special form of RBF*

Definition 1 (Symmetric Positive Definite (SPD) Matrix). A real $n \times n$ matrix A is symmetric positive definite if it is symmetric ($A = A^T$) and

$$x^T A x > 0 \quad \forall x.$$

A has a unique Cholesky factorization $A = R^T R$, where R is upper triangular with positive diagonal elements.

A has positive eigenvalues

Definition 2 (Maximum Likelihood Estimation (MLE)). We can either find the probability of the data given parameters, or find the parameters given the data. In the case of our Kriging models, we aim to estimate the parameters based on our limited data in order to create our surrogate.

Definition 3 (Stochastic Process). A model of a system that evolves over time or space in a random unpredictable manner.

Definition 4 (Covariance). A measure of the correlation between two or more sets of random variables

$$\text{cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)] \tag{1}$$

$$= E[XY] - \mu_x \mu_y \tag{2}$$

where μ_x and μ_y are the means of X and Y and E is the expectation. To make this make more sense, we can derive the more intuitive correlation from covariance by:

$$\text{cor}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} \tag{3}$$

(TODO) I still need to work on understanding what this means exactly.

3 2026-06-29 *Multi-Fidelity Analysis: Using multiple datasets for better approximations*

3.1 The general idea

In the event that we have multiple datasets describing a process, it is useful to combine data from an expensive function and a cheap approximation of said function. We want to sample the expensive function at a select few points that overlap with the sampling of the cheap function ($X_e \subset X_c$). Then we can try to correct points from Y_e with:

$$y_e = Z_p y_c + Z_d \quad (4)$$

3.2 Co-Kriging

From Forrester:

Here we use the auto-regressive model of Kennedy and O'Hagan (2000), which assumes that

$$\text{cov}\{Y_e(x_i), Y_c(x) \mid Y_c(x_i)\} = 0 \quad \forall x \neq x_i.$$

This means that no more can be learnt about $Y_e(x_i)$ from the cheaper code if the value of the expensive function at x_i is known (this is known as a *Markov property* which, in essence, says we assume that the expensive simulation is correct and any inaccuracies lie wholly in the cheaper simulation). (Forrester et al., 2008)

As in regular Kriging, we treat our expensive function as a stochastic process Z_e ,

$$Z_e(\mathbf{x}) = \rho Z_c(\mathbf{x}) + Z_d(\mathbf{x}) \quad (5)$$

Basically, we are modeling the expensive function as our cheap function multiplied by some correction factor ρ plus a separate difference model Z_d that estimates the difference between our cheap and expensive function at a point.

It helps me to see the final product and then dissect the parts. To use our co-kriging model to estimate a point $x^{(n_e+1)}$ we evaluate:

$$\hat{y}_e(x^{(n_e+1)}) = \hat{\mu} + \mathbf{c}^T \mathbf{C}^{-1} (\mathbf{y} - \mathbf{1}\hat{\mu}) \quad (6)$$

Where:

$\hat{\mu}$ the estimated global mean of the co-Kriging process

$\mathbf{c}$ a column vector of correlations between the new point $x^{(n_e+1)}$ and the observed points

$$\mathbf{c} = \begin{pmatrix} \hat{\rho}\hat{\sigma}^2\psi_c(\mathbf{X}_c, x^{(n+1)}) \\ \hat{\rho}^2\hat{\sigma}^2\psi_c(\mathbf{X}_e, x^{(n+1)}) + \hat{\sigma}^2\psi_d(\mathbf{X}_e, x^{(n+1)}) \end{pmatrix}$$

$\mathbf{C}$ the correlation matrix of the observed points

$$\mathbf{C} = \begin{pmatrix} \sigma^2\Psi_c(\mathbf{X}_c, \mathbf{X}_c) & \rho\sigma^2\Psi_c(\mathbf{X}_c, \mathbf{X}_e) \\ \rho\sigma_c^2\Psi_c(\mathbf{X}_e, \mathbf{X}_c) & \rho^2\sigma^2\Psi_c(\mathbf{X}_e, \mathbf{X}_e) + \sigma^2\Psi_d(\mathbf{X}_e, \mathbf{X}_e) \end{pmatrix}$$

$\mathbf{y}$ the vector of observed responses (cheap and expensive stacked)

$\mathbf{1}$ a column vector of ones, of the same length as $\mathbf{y}$

ψ or Ψ is a column vector or matrix calculated at each point with. Optimal theta values are found for each dimension and for both cheap points and expensive points (ψ_d)

$$\psi^{(i)} = \exp\left(-\sum_{j=1}^k \hat{\theta}_{j(\text{c or d})} \|x_j^{(n+1)} - x_j^{(i)}\|\right)$$

3.3 Co-Kriging – sampling plan revisited

From Forrester et al. (2007), our sampled points are:

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_c \\ \mathbf{X}_e \end{pmatrix} = (x_c^{(1)}, \dots, x_c^{(n_c)}, x_e^{(1)}, \dots, x_e^{(n_e)})^T \quad (7)$$

The difficulty is that

$$X_e \subset X_c$$

meaning we need to generate a random latin hypercube for our cheap points, and then pick the optimal subset of expensive points.

Thoughts. I think the easiest way to do this in code is to generate the optimal sampling plan for the subset, and then add points. The problem is that we need to be able to iterate over our sample set pairing one expensive sample point to one cheap sample point. I am trying to come up with an elegant solution that doesn't involve sorting the sample points in some messy way. Maybe I should just add points based on the Morris-Mitchell criterion one at a time? How can I pick a new point that maximizes distance from existing points? OK, I changed my mind, I think the optimal way to do this is finding a subset. If I try adding new points there is simply no way to ensure I have found the best plan. If I take a subset, I run into a similar problem. Since my expensive points must be a subset, I am not able to perturb the points, which means they will likely not be the best plan.

Here's what I came up with, I believe it is an accurate implementation of the exchange algorithm set out in Forrester et al. (2007) §4.

```
1
2
3 def find_subset(cube: np.ndarray, n: int, q: int = 2, iter: int = 5) ->
  np.ndarray:
4
5     rng = np.random.default_rng()
6     X_tracker = {
7         "phi": math.inf,
8         "X": np.empty([])
9     }
10
11     for _ in range(iter):
12         # restart to avoid local minima
13         X_best = rng.choice(cube, size=n, replace=False)
14         phi_best = mm_phiq(X_best, q)
15
16         if phi_best < X_tracker["phi"]:
17             X_tracker["phi"] = phi_best
18             X_tracker["X"] = X_best
19
20         for i in range(n):
21             # iterate over points in our subset
22
23             for point in cube:
24                 # try each point from larger sample set
25                 X_test = X_tracker["X"].copy()
26                 X_test[i] = point
27                 phi_best = mm_phiq(X_test, q)
28
29                 if phi_best < X_tracker["phi"]:
30                     X_tracker["phi"] = phi_best
```

```

31         X_tracker["X"] = X_test
32
33
34     return X_tracker["X"]
35
36
37 def align_subset(X_c: np.ndarray, X_e: np.ndarray) -> np.ndarray:
38
39     if X_c.shape[1] != X_e.shape[1]:
40         raise ValueError("X_c and X_e must have same number of
41                             dimensions")
42
43     rows_to_delete = []
44     for row in X_e:
45         rows_to_delete.append(np.where((X_c == row).all(axis=1))[0])
46
47     rows_to_delete = np.concatenate(rows_to_delete)
48
49     X_c = np.delete(X_c, rows_to_delete, axis=0)
50     X_tot = np.concatenate((X_e, X_c), axis=0)
51
52     return X_tot

```

`find_subset` works to find the best subsample of our original sampling plan. Then, `align_subset` deletes all the subsamples from our original sampling plan before prepending them back. This helps us pair the expensive sample points with their cheap counterparts later in the co-Kriging process.

4 Papers

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4.1 Guzman et al. (2026) — Toward Explaining GCR Modulation in a Dynamic Heliosheath

4.1.1 Definition Table

Heliosheath (HS) is the region of the heliosphere between the termination shock and the heliopause.

Unipolar region (UHS)

Sectored region (SHS)

Spatial effects relate to persistent plasma properties affecting particle transport over large areas. Examples include the heliospheric current sheet.

Temporal effects are transient modulation events and solar cycle variations advected outward from the Sun, generally lagging 1-2 yr from their creation to their arrival at the HS.

4.1.2 Abstract

GCR transport appears to be dominated by large, transient modulation structures advected from the Sun, while the role of spatially dictated diffusion is likely secondary. The stated goal of the paper is to find out the approximate relative contribution of each of these two factors.
hello

Open question 1. What is spatially dictated diffusion?

Answer (to Question 1). Spatially dictated diffusion refers to diffusion caused by persistent features like the heliospheric current sheet, which is a large-scale structure in the heliosphere that can influence the transport of GCRs.

4.1.3 Analytic Heliosphere Model

4.2 Puzzoni et al. (2026) — A new web-based tool for GCR modulation in the heliosphere

4.2.1 Abstract

This GCR tool is based on the Gleeson and Axford force field approximation solution to the parker transport equation and incorporates solar modulation via the correlation between GCRs and the mean sunspot number. This is a highly simplified model chosen for its fast evaluation speed and ease of use. This model is also validated against lots of empirical data sets including Voyager 1 and 2, and New Horizons.

Definition 5 (Parker Transport Equation). The Parker transport equation is a fundamental model in space physics. It describes how energetic charged particles move through the solar system's magnetized and turbulent plasma.

4.2.2 Model equations

As mentioned, this model is an implementation of the Gleeson and Axford solution to the parker transport equation. This solution is found when cosmic ray streaming flux is equal to zero.

$$\vec{S} = -\kappa \cdot \nabla f - \frac{\vec{U}}{3} \frac{\partial f}{\partial \ln p} = 0 \quad (8)$$

Where:

$\vec{S}$ Cosmic ray streaming flux. Set to zero for high energy GCRs

$\vec{U}$ Solar wind speed

$\frac{\partial f}{\partial \ln p}$ The momentum gradient. This term links the spatial and energy evolution of particles

∇f Gradient of the phase space distribution function

Again, this solution is only valid when cosmic ray streaming flux $\vec{S} = 0$, which is only true for high energy GCRs. Additionally, the model assumes a spherically symmetric, steady state heliosphere centered on the Sun, which is of course, not realistic.

Solving Eq. 8, one obtains for f

$$f(r, E) = f(R_{HP}, E + \phi) \quad (9)$$

Where:

R_{HP} is the radial distance to the Heliopause

E is the kinetic energy of the GCRs

ϕ is the modulation potential

ϕ is found by

$$\phi = \int_r^{R_{HP}} \frac{U(r')}{3g(r')} dr' \quad (10)$$

In the above, U is still solar wind speed, and $g(r')$ is the radial dependence of the spatial diffusion coefficient, which has the form $k(r, p) = g(r)wp$, p again being momentum, and w being particle velocity. This modulation potential is where all the tunable parameters exist.

4.3 M. C. Kennedy and A. O'Hagan (2000) — Predicting the output from a complex computer code

4.4 Forrester et al. (2007) — Multi-fidelity optimization via surrogate modelling